

One-Day-Ahead Value-at-Risk Forecasting for SPY

A Frozen Out-of-Sample Study of Historical Simulation, GARCH-t, HAR Quantile Regression, and Small Neural Quantile Models

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1 Executive Summary

This report studies one-day-ahead Value-at-Risk (VaR) forecasting for SPY log returns at the 1%, 5% and 10% lower tails, under a strictly causal rolling-window protocol. Six model families are compared on a frozen out-of-sample period starting 2008-01-01: rolling Historical Simulation (HS), GARCH(1,1) with Student- t innovations, GJR-GARCH(1,1,1)- t , linear (HAR-style) quantile regression, a small multi-quantile MLP, and a small GRU quantile model. Feature ablations isolate the incremental value of realized volatility (rv5), bipower variation (bv), and jump/downside-asymmetry information. All numbers in this report are regenerated by scripts from machine-readable prediction artifacts; none are hand-entered.

2 Problem Definition

For each forecast origin t we predict the conditional return quantile $q_{\alpha,t+1}$ with $P(r_{t+1} \leq q_{\alpha,t+1} \mid \mathcal{F}_t) = \alpha$ for $\alpha \in \{0.01, 0.05, 0.10\}$, using only information available at the end of day t . A violation is defined as $r_{t+1} \leq \hat{q}_{\alpha,t+1}$ (VaR is the return quantile itself; lower-tail values are typically negative).

3 Data

Daily SPY observations with columns `log_ret`, `rv5` (5-minute realized variance) and `bv` (bipower variation). See Appendix A for the machine-generated data audit.

4 Exploratory Analysis

See Figures 1–2. Key features: volatility clustering, heavy tails (excess kurtosis), strong persistence of log realized measures, and a positive link between negative returns and next-day volatility.

5 Experimental Protocol

Development period: all dates before 2008-01-01 (rolling-origin validation over 2005–2007, with a documented slight adjustment for the 1500-day window whose earliest feasible origin is late 2005). Final test: all dates from 2008-01-01. Candidate windows: 1000 and 1500 trading days only; the primary window is chosen from development evidence (see Section 10). All models share the same

forecast dates. The final configuration and data hash are frozen in `docs/FREEZE_MANIFEST.md` before any final-test run; the gate is enforced by tests and scripts.

6 Models

6.1 M0 – Historical Simulation

The empirical quantile of the rolling-window returns: $\hat{q}_\alpha = \hat{F}_t^{-1}(\alpha)$ over $\{r_{t-W+1}, \dots, r_t\}$. Non-parametric benchmark; implicitly assumes a stationary return distribution within the window, so regime shifts are absorbed only with a lag.

6.2 M1 – GARCH(1,1)-Student- t

Classical conditional volatility model with conditional variance

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \varepsilon_t = \sigma_t z_t, \quad (1)$$

where z_t is an i.i.d. unit-variance standardized Student- t innovation with ν degrees of freedom (variance 1, so that $\text{Var}(\varepsilon_t | \mathcal{F}_{t-1}) = \sigma_t^2$). The one-day-ahead VaR is

$$\text{VaR}_{\alpha,t+1} = \mu + F_t^{-1}(\alpha; \nu) \hat{\sigma}_{t+1}, \quad (2)$$

with $F_t^{-1}(\alpha; \nu)$ the quantile of the standardized Student- t (implemented via `arch.univariate.StudentsT`, equal to $t_\nu^{-1}(\alpha) \sqrt{(\nu-2)/\nu}$); μ is a constant mean. Parameters are re-estimated by MLE on every rolling window.

6.3 M4 – GJR-GARCH(1,1,1)-Student- t

Adds a leverage term to the variance recursion,

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (3)$$

capturing asymmetric responses of volatility to negative shocks; VaR construction identical to M1. Serves as the downside-asymmetry robustness extension.

6.4 M2 – Linear / HAR Quantile Regression

Direct conditional quantile estimation by minimizing the pinball loss,

$$\hat{\beta}_\tau = \arg \min_{\beta} \sum_t \rho_\tau(y_{t+1} - x_t^\top \beta), \quad \rho_\tau(u) = u(\tau - I(u < 0)), \quad (4)$$

fit separately for $\tau \in \{0.01, 0.05, 0.10\}$ on the same feature sets as the MLP (so the linear-vs-nonlinear contrast is not confounded by information). Independent fits may cross; the crossing rate is reported.

6.5 M3 – Multi-Quantile MLP

A small MLP with joint pinball loss over the three tails and a structurally non-crossing head:

$$q_{0.01} = z_1, \quad q_{0.05} = z_1 + \text{softplus}(z_2), \quad q_{0.10} = q_{0.05} + \text{softplus}(z_3), \quad (5)$$

so that $q_{0.01} \leq q_{0.05} \leq q_{0.10}$ holds by construction ($\text{softplus} > 0$). Targets are standardized with train-only statistics inside each rolling window and the affine transform is inverted on output (Scheme B, audit fix). Training uses early stopping on the last 10% of the window in time order; scalars are fitted on training rows only.

6.6 M5 – GRU Quantile

A one-layer GRU over the last 22 daily feature vectors with the same ordered head and loss. The role is a robustness extension: it tests whether an explicit sequence representation adds value beyond the structured HAR-style features already present, not a substitute for the MLP. Parameters are deliberately kept small.

6.7 Gaussian-GARCH diagnostic

GARCH(1,1) with Gaussian innovations is estimated only in development as a diagnostic reference.

7 Feature Engineering

Information sets (identical for M2 and M3, so Linear-vs-MLP differences are attributable to the mapping rather than the inputs): **F0** returns only (lags 1/2/5/22, abs, squared); **F1** adds $\log rv5$, $\sqrt{rv5}$, 5-day and 22-day HAR aggregations of $\log rv5$; **F2** adds the same block for bv ; **F3** adds jump proxy $\max(rv5 - bv, 0)$, relative jump, and a 5-day downside-shock intensity. All features are causal by construction and verified by truncation-invariance tests.

8 Rolling Forecast Design

A unified rolling engine controls window boundaries, feature cutoffs, preprocessing (scalers fitted on the window training rows only), model refits (daily full refit for every model), and prediction recording. Early stopping for neural models uses the last 10% of the window in time order. All outputs share one schema (RESULTS_SCHEMA.md) with `fit_status` logging; failures are never silently dropped.

9 VaR Backtesting Methodology

Per model and tail: empirical failure rate, Kupiec unconditional coverage test, Christoffersen independence and conditional-coverage tests, mean pinball loss, quantile-crossing rate, violation clustering, Dynamic Quantile test (reported as an additional diagnostic), Diebold–Mariano tests and moving-block bootstrap for pairwise loss comparisons. Finite-sample caveats apply: the 1% tail has very few expected violations even in a decade-long test, so coverage tests have low power and small numerical differences must not be over-read.

10 Development / Validation Results

Window comparison tables: `outputs/development/window_selection.csv` and `window_selection_cells.csv`. Decision rule (predeclared, audit-fixed aggregation): per (model, feature-set, tail) cell the candidate-relative normalized loss (cell loss divided by the cell mean across the two candidates) is averaged with equal weight across all cells; per-cell winners, pairwise win counts and drop-one sensitivity are reported alongside. If the equal-weight aggregate and the raw pinball-sum aggregate disagree, the longer window is chosen conservatively for 1% tail stability. The longer window provides more effective tail observations for the 1% quantile (about 15 vs 10 expected violations) and hence more stable empirical quantiles, at the cost of slower regime adaptation; the short window adapts faster. The primary window selected from development evidence is 1500.

11 Frozen Out-of-Sample Results

All models evaluated on 2640 identical forecast dates ($\geq 2008-01-01$). Primary seed 42. Table 1 reports coverage and loss statistics.

Table 1: Frozen out-of-sample backtesting results by model and tail.

Tail 1%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	41	0.0155	0.0082	0.0000	0.0000	0.0006
GARCH-t	2640	20	0.0076	0.1909	0.5805	0.3650	0.0004
LinQR	2640	57	0.0216	0.0000	0.0411	0.0000	0.0005
MLP	2640	26	0.0098	0.9375	0.0000	0.0000	0.0010
GJR-t	2640	19	0.0072	0.1276	0.5996	0.2729	0.0004
GRU	2640	4	0.0015	0.0000	0.0031	0.0000	0.0005
Tail 5%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	140	0.0530	0.4791	0.0000	0.0002	0.0017
GARCH-t	2640	113	0.0428	0.0822	0.6821	0.2031	0.0013
LinQR	2640	161	0.0610	0.0122	0.0020	0.0004	0.0013
MLP	2640	178	0.0674	0.0001	0.0021	0.0000	0.0026
GJR-t	2640	112	0.0424	0.0670	0.3700	0.1250	0.0013
GRU	2640	120	0.0455	0.2768	0.1428	0.1892	0.0014
Tail 10%							
Model	N	Viol	Rate	Kupiec p	Ind p	CC p	Pinball
HS	2640	254	0.0962	0.5141	0.0005	0.0020	0.0024
GARCH-t	2640	215	0.0814	0.0011	0.1316	0.0015	0.0021
LinQR	2640	297	0.1125	0.0354	0.1047	0.0294	0.0021
MLP	2640	335	0.1269	0.0000	0.0698	0.0000	0.0042
GJR-t	2640	217	0.0822	0.0017	0.1124	0.0021	0.0021
GRU	2640	318	0.1205	0.0007	0.0247	0.0002	0.0023

12 Tail-Level Model Comparison

See Figure 7. Rankings are tail-specific; a single universal winner is not assumed.

13 Feature Ablations

Table 2 and Figure 8. Ablation interpretation is confined to M2/M3 across F0–F3.

14 Crisis and Regime Analysis

Figure 9 and Table 3 stratify frozen results into predefined regimes: 2008–2009 crisis, 2010–2012 elevated volatility, 2013–2014 calm, 2015–2016 stress, 2017 calm, 2018 spike. This is an explanatory analysis of frozen results, not a basis for re-optimization.

Table 2: Feature ablation: failure rate and mean pinball by information set.

Model	Tail	Rate	Pinball
M2-F0	1%	0.0170	0.0004
M2-F0	5%	0.0542	0.0014
M2-F0	10%	0.1034	0.0022
M2-F1	1%	0.0174	0.0004
M2-F1	5%	0.0557	0.0013
M2-F1	10%	0.1080	0.0021
M2-F2	1%	0.0216	0.0005
M2-F2	5%	0.0606	0.0013
M2-F2	10%	0.1121	0.0021
M2-F3	1%	0.0216	0.0005
M2-F3	5%	0.0610	0.0013
M2-F3	10%	0.1125	0.0021
M3-F0	1%	0.0083	0.0018
M3-F0	5%	0.0553	0.0039
M3-F0	10%	0.1208	0.0064
M3-F1	1%	0.0076	0.0016
M3-F1	5%	0.0663	0.0033
M3-F1	10%	0.1212	0.0052
M3-F2	1%	0.0068	0.0017
M3-F2	5%	0.0807	0.0034
M3-F2	10%	0.1269	0.0046
M3-F3	1%	0.0098	0.0010
M3-F3	5%	0.0674	0.0026
M3-F3	10%	0.1269	0.0042

15 Robustness

Seed robustness for the neural models (predeclared seeds 7, 42, 2026; headline series is the primary seed 42, not an ensemble): `outputs/tables/seed_robustness.csv`. DM and block-bootstrap pairwise comparisons: Table 4.

16 Limitations

- The 1% tail has few expected violations (≈ 2.6 per year); coverage tests have low power.
- Neural models are retrained daily on at most 1500 observations; capacity is deliberately small.
- rv5/bv are daily aggregates; intraday dynamics are not modeled.
- DM/block-bootstrap inference at extreme tails relies on asymptotic approximations; results are interpreted as indicative.

17 Conclusion

The conclusion is drawn from the frozen out-of-sample evidence in Section 11 and the analysis sections.

Table 3: Failure rate by regime (primary configuration per model).

Regime	GARCH-t	GJR-t	GRU	HS	LinQR	MLP
calm_2013_2014	0.0417	0.0417	0.0317	0.0020	0.0575	0.0714
calm_2017	0.0159	0.0159	0.0279	0.0120	0.0239	0.0757
crisis_2008_2009	0.0635	0.0635	0.0615	0.1567	0.0774	0.0813
elevated_2010_2012	0.0424	0.0451	0.0517	0.0345	0.0663	0.0557
spike_2018	0.0569	0.0488	0.0569	0.1057	0.0976	0.0976
stress_2015_2016	0.0337	0.0298	0.0397	0.0357	0.0496	0.0556

At the 1% tail, the model with the empirical failure rate closest to the nominal level is MLP (0.0098 vs target 0.01); the lowest mean pinball loss is achieved by GJR-t (0.00035). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

At the 5% tail, the model with the empirical failure rate closest to the nominal level is HS (0.0530 vs target 0.05); the lowest mean pinball loss is achieved by GJR-t (0.00126). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

At the 10% tail, the model with the empirical failure rate closest to the nominal level is HS (0.0962 vs target 0.10); the lowest mean pinball loss is achieved by GJR-t (0.00205). Closeness of the unconditional failure frequency is not, by itself, evidence of adequate conditional VaR dynamics (see the coverage tests and the violation-clustering diagnostics below).

The GARCH family (GARCH-t, GJR-t) achieves the lowest mean pinball loss (sharpness) at all three tails and strong 1% tail calibration (failure rates 0.0428 / 0.0424 vs 0.0530 for HS), with no evidence of violation clustering at the 1% and 5% tails. Two caveats: at the 5% and 10% tails HS is marginally closer to the nominal frequency (5% tail 0.0530 vs 0.0428), and the 10% conditional-coverage test is rejected for the GARCH family (CC p 0.0015 / 0.0021); crisis-period 5% failure rates are 0.1567 (HS) vs 0.0635 (GARCH-t), documenting slow regime adaptation of historical simulation.

The neural models do not outperform the classical baselines on this frozen panel: the Holm-corrected DM p-values at the 5% tail are 0.0000 (GARCH-t vs MLP) and 0.0000 (LinQR vs MLP); the GRU is strongly over-conservative at the 1% tail (failure rate 0.0015). Across-seed dispersion is reported in the robustness section rather than summarized qualitatively here. This is a valid negative result on this information set and window.

Feature ablations (F0 returns -; F3 full block) change mean pinball at the 5% tail by LinQR: 3.5%, MLP: 33.0%; the realized-measure block adds information for the linear model and the MLP, but it does not change the model ranking. The F3 effect is attributed to the jump/downside feature block as a whole, not to any single component.

18 References

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A Machine-Generated Data Audit

docs/DATA_AUDIT.md (generated by `scripts/audit_data.py`).

B Artifacts

All prediction panels: `outputs/predictions/*.parquet` with sidecar manifests; metric tables: `outputs/tables/*.csv`; figures: `outputs/figures/*.png`. Freeze manifest: `docs/FREEZE_MANIFEST.md`.

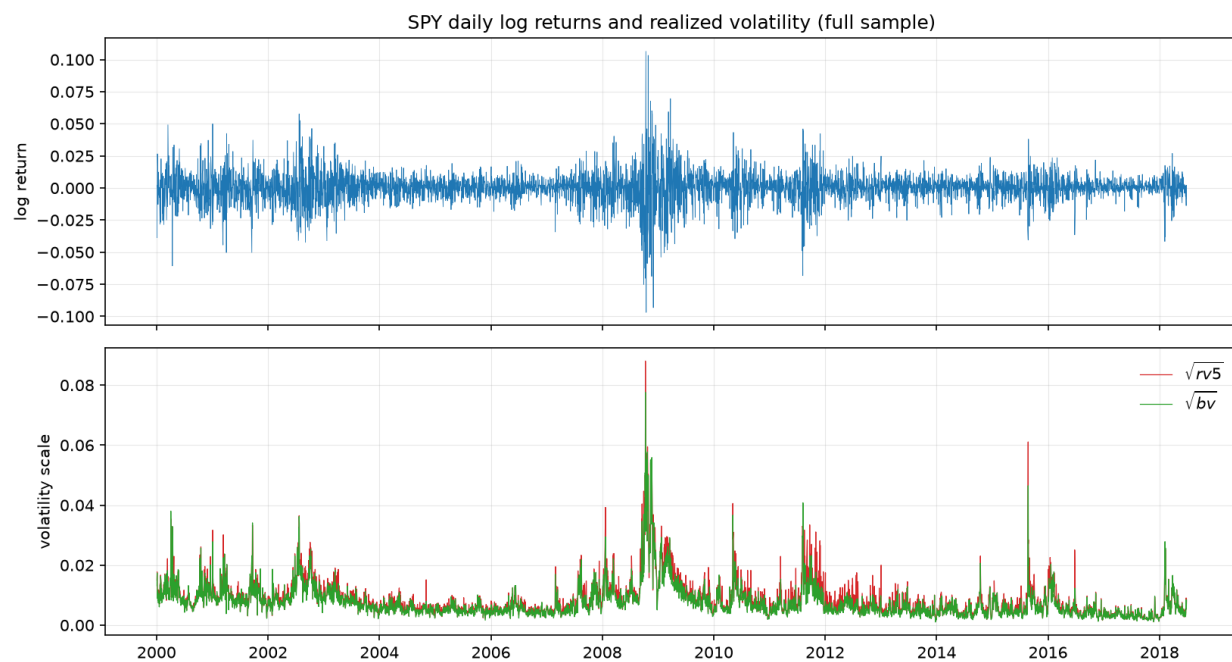


Figure 1: SPY log returns and volatility scale, full sample.

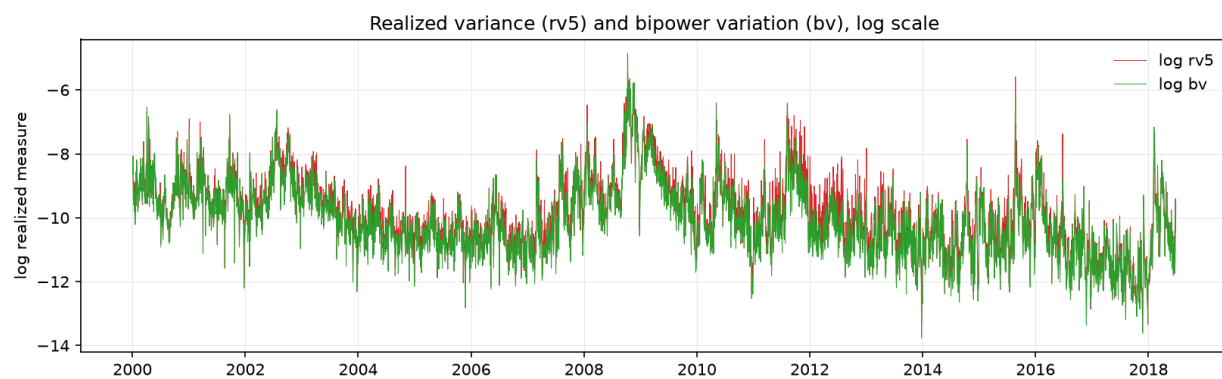


Figure 2: Realized variance and bipower variation (log scale).

Table 4: Pairwise loss comparisons (negative DM favors model_a).

Pair	Tail	DM	DM p	Boot p	Favors	Headline
M0 vs M1	1%	3.7684	0.0002	0.1940	b	
M0 vs M2	1%	2.0038	0.0451	0.1170	b	
M0 vs M3	1%	-4.5745	0.0000	0.1300	a	
M0 vs M4	1%	3.9258	0.0001	0.1810	b	
M0 vs M5	1%	1.2483	0.2119	0.5600	b	
M1 vs M2	1%	-1.9093	0.0562	0.3800	a	
M1 vs M3	1%	-5.0865	0.0000	0.1530	a	
M1 vs M4	1%	2.0895	0.0367	0.0670	b	
M1 vs M5	1%	-9.8714	0.0000	0.0010	a	
M2 vs M3	1%	-4.7918	0.0000	0.1150	a	
M2 vs M4	1%	2.0636	0.0391	0.3560	b	
M2 vs M5	1%	-0.1646	0.8693	0.9060	a	
M3 vs M4	1%	5.1678	0.0000	0.1490	b	
M3 vs M5	1%	3.9057	0.0001	0.2100	b	
M4 vs M5	1%	-10.8882	0.0000	0.0010	a	
M0 vs M1	5%	4.3445	0.0000	0.0990	b	
M0 vs M2	5%	4.7912	0.0000	0.0380	b	
M0 vs M3	5%	-3.9424	0.0001	0.1210	a	
M0 vs M4	5%	4.5717	0.0000	0.0710	b	
M0 vs M5	5%	3.5849	0.0003	0.0700	b	
M1 vs M2	5%	-0.8104	0.4177	0.5680	a	
M1 vs M3	5%	-4.7458	0.0000	0.1050	a	
M1 vs M4	5%	3.0994	0.0019	0.0070	b	
M1 vs M5	5%	-1.5826	0.1135	0.1950	a	
M2 vs M3	5%	-4.7969	0.0000	0.1000	a	
M2 vs M4	5%	1.5818	0.1137	0.3280	b	
M2 vs M5	5%	-0.7365	0.4614	0.4930	a	
M3 vs M4	5%	4.8545	0.0000	0.1010	b	
M3 vs M5	5%	4.7181	0.0000	0.1060	b	
M4 vs M5	5%	-2.3117	0.0208	0.0920	a	
M0 vs M1	10%	4.3293	0.0000	0.0600	b	
M0 vs M2	10%	4.9103	0.0000	0.0300	b	
M0 vs M3	10%	-4.2991	0.0000	0.1100	a	
M0 vs M4	10%	4.5523	0.0000	0.0410	b	
M0 vs M5	10%	2.1835	0.0290	0.1100	b	
M1 vs M2	10%	0.1414	0.8876	0.9130	b	
M1 vs M3	10%	-4.7636	0.0000	0.0970	a	
M1 vs M4	10%	2.9749	0.0029	0.0060	b	
M1 vs M5	10%	-2.8651	0.0042	0.0510	a	
M2 vs M3	10%	-4.8492	0.0000	0.0970	a	
M2 vs M4	10%	0.9425	0.4578	0.5390	b	
M2 vs M5	10%	-3.3584	0.0008	0.0170	a	
M3 vs M4	10%	4.8232	0.0000	0.0960	b	
M3 vs M5	10%	4.6733	0.0000	0.1020	b	

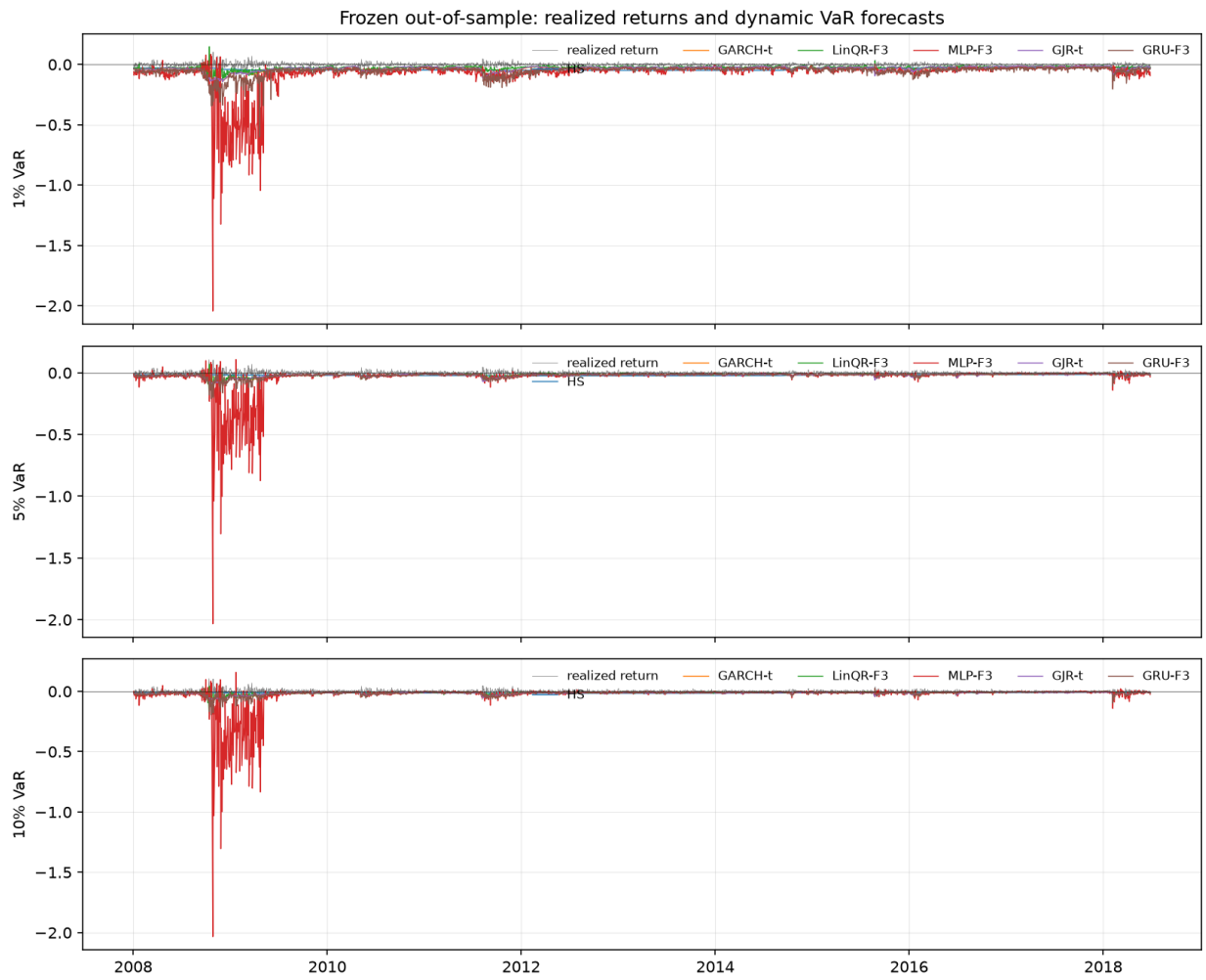


Figure 3: Frozen out-of-sample realized returns with dynamic VaR forecasts.

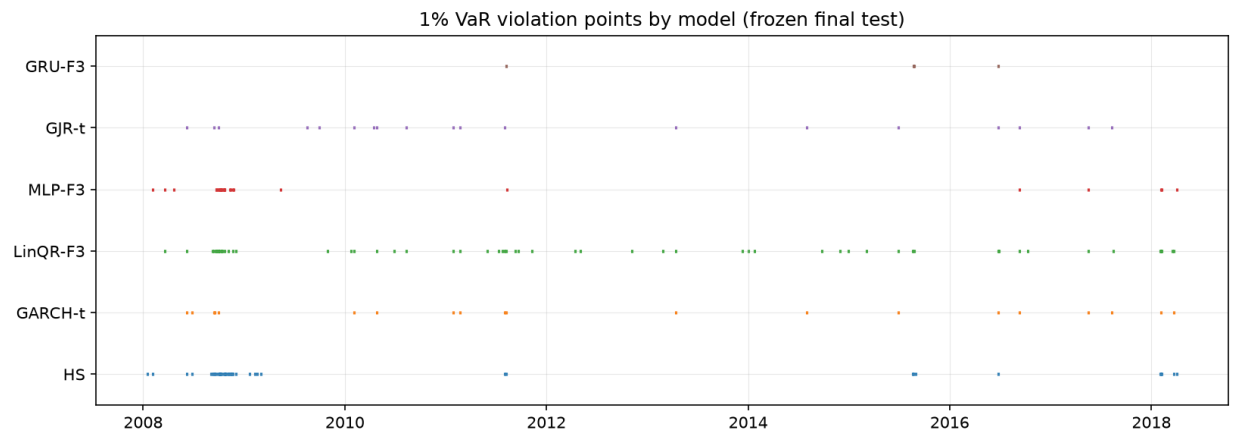


Figure 4: 1% VaR violation points by model.

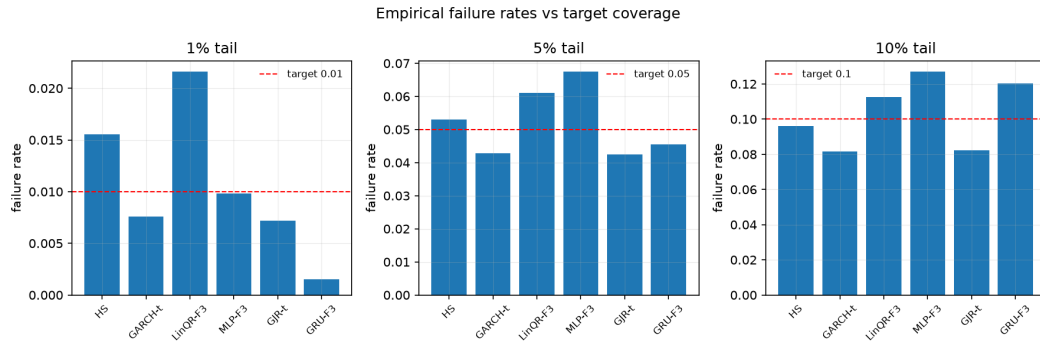


Figure 5: Empirical failure rates vs target coverage.

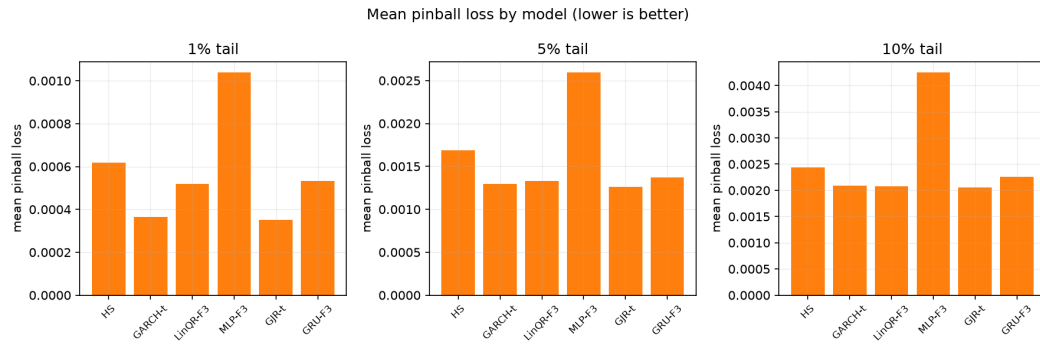


Figure 6: Mean pinball loss by model and tail.

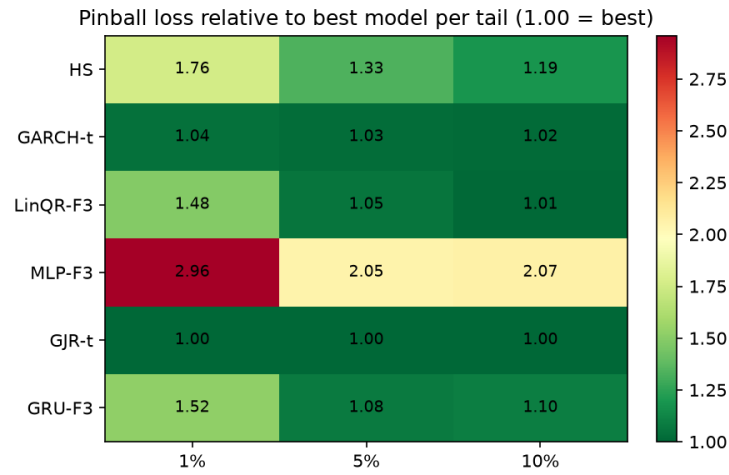


Figure 7: Pinball loss relative to best model per tail.

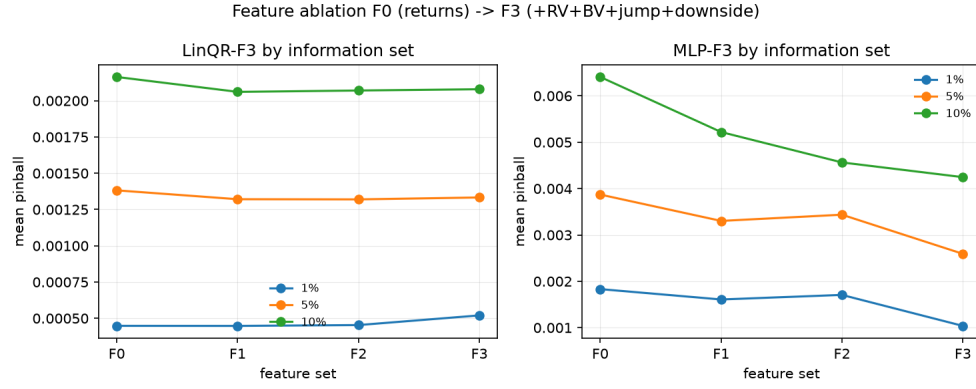


Figure 8: Feature ablation F0–F3 for linear quantile regression and MLP.

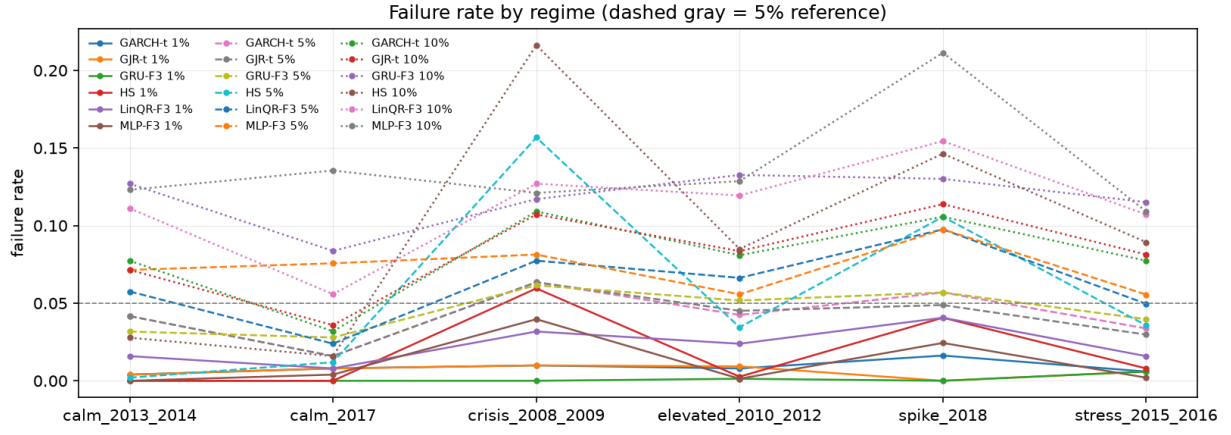


Figure 9: Failure rate by regime.

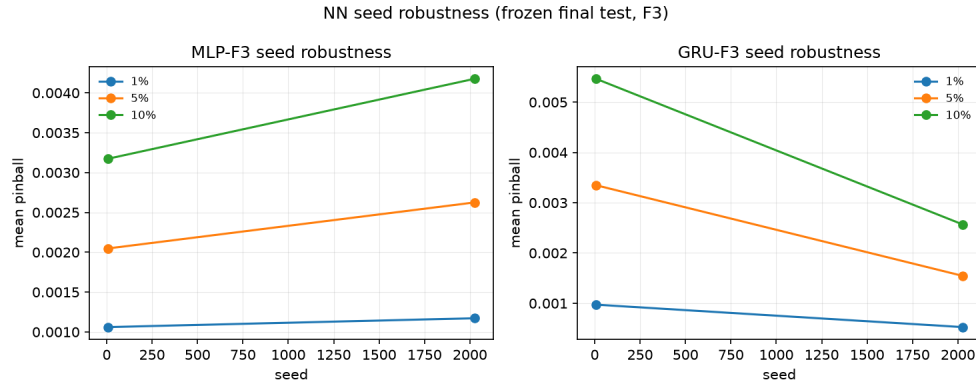


Figure 10: Neural seed robustness.